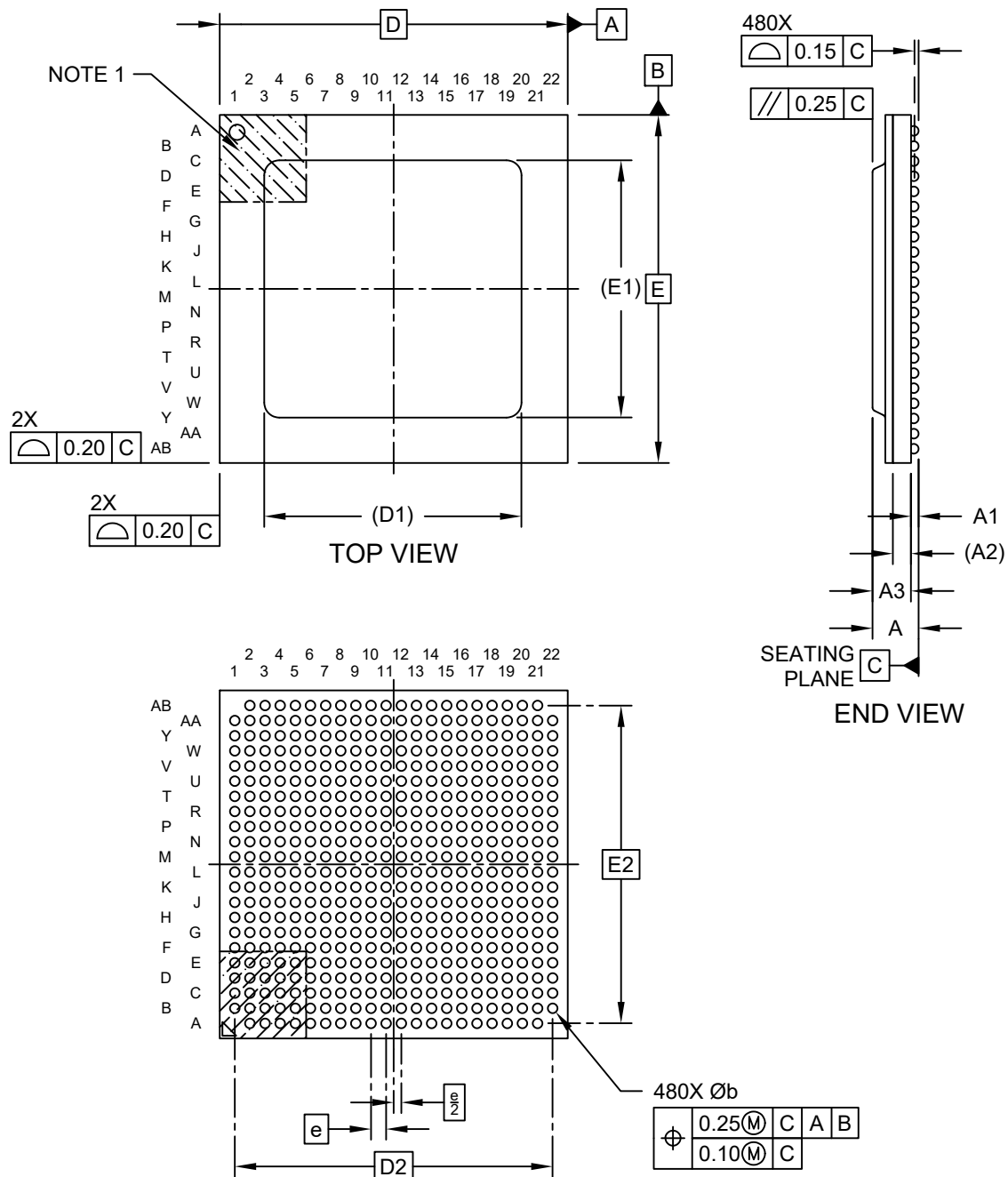


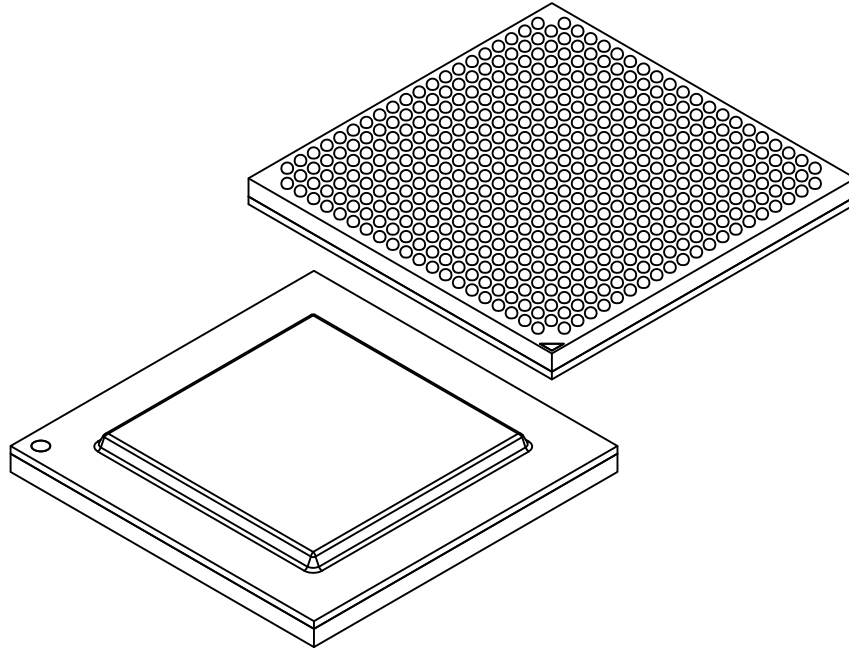
480-Ball Thick Ball Grid Array (CPC) - 23x23x3.32 mm Body [BBGA] Flip Chip; Microsemi Legacy Package

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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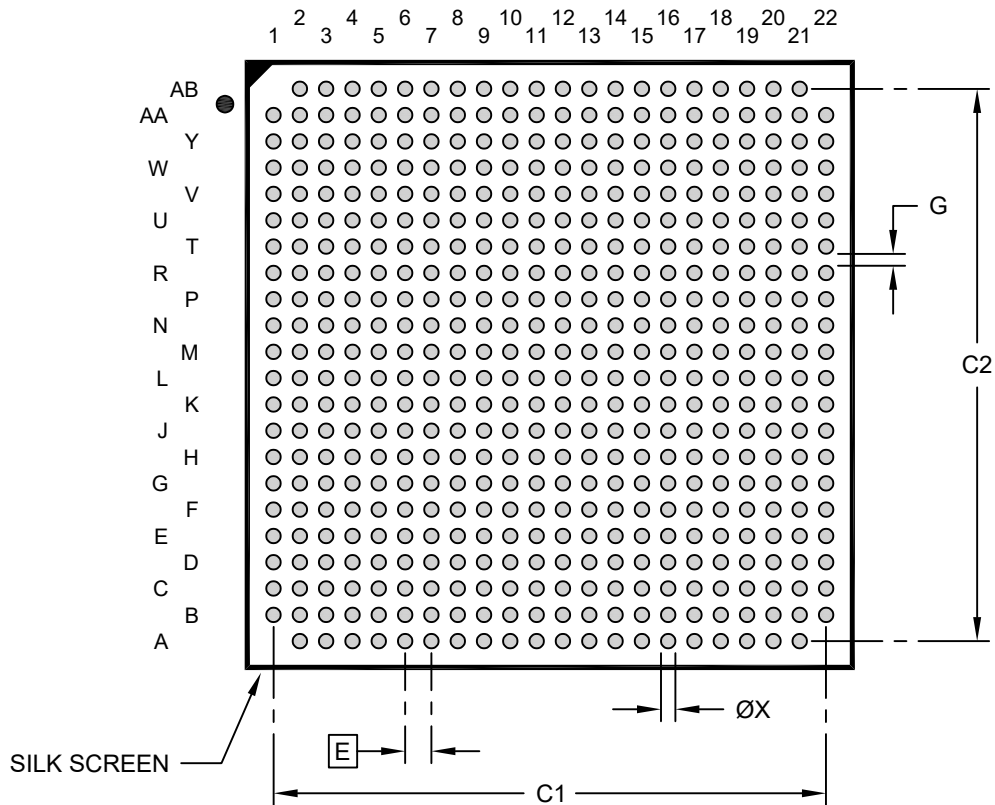
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		480		
Pitch	e		1.00 BSC		
Overall Height	A		2.76	3.04	3.32
Ball Height	A1		0.40	0.50	0.60
Substrate Thickness	A2		1.20 REF		
Package Thickness	A3		2.36	2.54	2.72
Overall Length	D		23.00 BSC		
Cap Length	D1		17.00 BSC		
Ball Array Length	D2		21.00 REF		
Overall Width	E		23.00 BSC		
Cap Width	E1		17.00 BSC		
Ball Array Width	E2		21.00 REF		
Ball Diameter	b		0.50	0.64	0.70

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

480-Ball Thick Ball Grid Array (CPC) - 23x23x3.32 mm Body [BBGA] Flip Chip; Microsemi Legacy Package

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	1.00 BSC		
Contact Pad Spacing	C1		21.00	
Contact Pad Spacing	C2		21.00	
Contact Pad to Contact Pad	G	0.45		
Contact Pad Diameter	X			0.55

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.